

Construction-pure (R) and (L)

The energy premises as structure theorems of the fold, and the kills that actually fire

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Abstract

Round 391 tests the suspicion that the two energy premises of COMPOSE⁻ — (R) : $B(\omega + \beta, \omega + \beta) \leq KD$ and (L) : $L_1 \leq Cm^\gamma$ with $\gamma < 1/4$ — are theorems of the fold construction, not of the arithmetic of the weights. Scramble (r383/r386) breaks neither. What is proved: the Euclidean split $\|\Sigma\|^2 + \|\Delta\|^2 = 2(\|P_\beta\|^2 + \|P_\omega\|^2)$ and the correlation identity $\|\Sigma\|^2/\|\Delta\|^2 = (1 + \eta)/(1 - \eta)$; Cauchy–Schwarz counting $L_1^2 \leq mD$; Fejér Gram; the DC Rayleigh $R = 3.6875$ on the length-16 all-ones toy. What is *not* proved: a cofinal K via block-Gershgorin (fully diagonally dominant on only 4/42 of the core ladder; atom-Gershgorin is $\lambda \leq H$, already too coarse), or $\gamma < 1/4$ from counting (the $L_1^2 \leq mD$ bridge is tautological with $n_{\text{eff}} \sim m$). Weight randomisation at frozen fold geometry respects $K = 4$ except under positive alignment $P_\omega \approx P_\beta$. The geometry kill is DC/sign-flatten, *not* merged blocks. No RH claim.

Status. Lemmas 1–3 are proved (finite algebra). Lemma 4 is the trivial envelope, too coarse. Lemma 5 is a census of the white-block class. Proposition 1 records (R) reduced. Proposition 2 records (L) reduced. Proposition 3 names the COMPOSE⁻ remainder. No RH claim.

1 Recollection

Write $\Sigma = \omega + \beta = P_\omega + P_\beta$, $\Delta = \beta - \omega = P_\beta - P_\omega$, $D = \|\Delta\|^2$, $L_1 = \|\Delta\|_1$, and B the positional Fejér form of [3, 1]. The named remainder of r386 is $B(\Sigma, \Sigma) \leq KD$. COMPOSE⁻ of [3] still asks (R) , (L) , living (Z') [1], $M_3 \leq \phi(m)$, and the r263 dictionary. Claim boundary: finite identities and named reductions. Not evidence for or against the Riemann hypothesis.

2 Shared machinery

Lemma 1 (Euclidean split). $\|\Sigma\|^2 + \|\Delta\|^2 = 2(\|P_\beta\|^2 + \|P_\omega\|^2)$. Writing $\eta = 2\langle P_\beta, P_\omega \rangle / (\|P_\beta\|^2 + \|P_\omega\|^2)$ one has $\|\Sigma\|^2/\|\Delta\|^2 = (1 + \eta)/(1 - \eta)$ whenever $|\eta| < 1$. Consequently $B(\Sigma, \Sigma)/D = \text{ray}(\Sigma) \cdot (1 + \eta)/(1 - \eta)$ with $\text{ray}(\Sigma) = B(\Sigma, \Sigma)/\|\Sigma\|^2$.

Lemma 2 (Fejér Gram). B is a Gram form. Atom-Gershgorin of the Fejér kernel gives $\lambda_{\max}(K) \leq H$, hence $B(u, u) \leq H\|u\|^2$, already recorded in [2] and too coarse for an absolute K .

Lemma 3 (Cauchy–Schwarz counting). $L_1^2 \leq mD$ for every real block field of length m .

Lemma 4 (Atom-Gershgorin is H). The Fejér Toeplitz matrix has diagonal 1 and off-diagonal row-sum $H - 1$. Gershgorin disks give $\lambda \leq H$. This is Lemma 2.

On a length-16 all-ones vector with $H = 4$ one has $R = 3.6875 > C_{\text{MAIN}} = 3/10$: a field-independent bound $C_{\text{MAIN}} \leq 1$ remains refuted. Exact period-2 gives $R = 1/16$ (anti-alignment kills the energy).

Lemma 5 (White-block class, census). *On the 42-rung core, $\text{med } \|\Sigma\|^2/D = 0.999$ and $\text{med } \cos(P_\beta, P_\omega) = -0.0006$; $\text{ray}(\Sigma) \leq 3.099 < 4$. Block length has median 4 and maximum 13. Superblock Gershgorin of Σ is fully diagonally dominant on only 4/42 (minimum fraction 0.667).*

3 Premise (R)

Proposition 1 ((R) , reduced). *$B(\Sigma, \Sigma) \leq 4D$ on the construction class: CORE-42 (0/42 violations, maximum 2.8196 at kz19), χ_3 -42 (0/42, maximum 2.746 at kz67), and the R -star kz37 ($B_\Sigma/D = 2.975 < 4$). Independent weight randomisation at frozen fold geometry (permute, Gauss, Rademacher, exponential, Dirichlet) stays below $K = 4$. Positive alignment $P_\omega = aP_\beta$ with $a = 0.3$ already exceeds 4 and $a \rightarrow 1$ is unbounded — there is no field-independent K . DC-flatten and sign-kill break $K = 4$; merged folds and shuffled blocks do not. Superblock Gershgorin is refuted as a SATZ. The mutant $K/2 = 2$ fails on 9/42 and at kz37.*

Outcome: **REDUCED** to the white-block class ($\eta \approx 0$, $\text{ray} = O(1)$), with $K = 4$ a census envelope. Remainder: prove $\text{ray}(\Sigma) \cdot (1 + \eta)/(1 - \eta) \leq K$ from fold pairing, not Gershgorin.

4 Premise (L)

The von Mangoldt triangle on positions still has slope $+0.307 \not< 1/4$ [2]. The new counting is at block level. Lemma 3 is a theorem, and on the core it is nearly sharp ($L_1/\sqrt{mD}$ median 0.702) because $n_{\text{eff}} \sim m$ is the r301 identity $n_{\text{act}} = m$. The translation $\gamma_{\text{CS}} = \frac{1}{2} + \frac{1}{2}\text{sl}_D$ equals $+0.214 < 1/4$ on CORE-42 and $+0.192$ on χ_3 , but this is the D -decay of the L2 cascade, not an independent construction bound. Measured slope(L_1) = $+0.202$ (CORE-42) and $+0.182$ (χ_3) stays below $1/4$ under weight randomisation; the DC mutant $L_1 = m$ has slope 1 and kills every field-independent $\gamma < 1/4$.

Proposition 2 ((L) , reduced). *L_1 is construction-class: scramble and independent weight-rand leave $L_1 \leq 3$ on the pins; the CORE-42 log-log slope is $+0.202 < 1/4$. Cauchy-Schwarz counting is a SATZ and is tautological with $n_{\text{eff}} \sim m$. The triangle does not close. The geometry kill is the DC field $L_1 = m$. A cofinal $\gamma < 1/4$ is not proved.*

Outcome: **REDUCED**. Remainder: a construction bound on the bulk defect mass (equivalently, cofinal $\gamma < 1/4$).

5 Consequence

Proposition 3 (COMPOSE[−] after r391). *(R) and (L) are construction-class censuses, not cofinal SATZ. COMPOSE[−] still requires*

(Z') living ladder $\wedge M_3 \leq \phi(m) \wedge \text{Dict} \wedge$ the two envelopes $K = 4$, $\Lambda = 3$ (or the named remainders).

The r389 Weyl energy owns the signed cancellation objects (assist, C_ε , Z_{loc}), not the positive quantities S_F , D , L_1 .

6 Machine check

`verify_construction_rl.py` and `construction_pure_rl_probe.py` gate the toys (DC, period-2, CS, counting, split, Gershgorin, $\gamma = 1/4$), the $w = 9$ white-block pins, scramble non-break, kz37, weight-rand, DC/align/merge mutants, and on `--full` the CORE-42+ χ_3 -42 census (the 181-pack and FRAME-B exceeded two minutes and were skipped, disclosed). Final line CONSTRUCTION PURE RL VERIFIED. No RH claim.

References

- [1] S. Hamann, *Living-ladder (Z') and cofinal R_0* , `rh/problem/compose_premises2.pdf`, 2026.
- [2] S. Hamann, *The compose premises (R), (L), (Z)*, `rh/problem/compose_premises.pdf`, 2026.
- [3] S. Hamann, *The compose lemma, reduced*, `rh/problem/compose_lemma.pdf`, 2026.
- [4] S. Hamann, *The Weyl energy of cancellation, reduced*, `rh/problem/weyl_energy.pdf`, 2026.